STATISTICS PRACTICALS — SHORT NOTES

Formulas | Steps | Methods | Charts

1. Histogram, Average & Variance

Steps:

1. List all data values (xi)
2. Compute mean: x■ = Σxi / n
3. Find deviations: xi – x■
4. Square deviations: (xi–x■)²
5. Sum squared deviations
6. Create bin intervals (class widths)
7. Count freq per bin → Histogram

Formulas:

x■ = (1/n) · Σxi
σ² = Σ(xi–x■)² / n [Population]
s² = Σ(xi–x■)² / (n–1) [Sample]
σ = √σ² | s = √s²
Excel: AVERAGE(), STDEV(), VAR()

Chart: Column / Bar Histogram

2. Stem & Leaf Plot

Steps:

1. Sort data in ascending order
2. Identify stems (tens digit)
3. Identify leaves (units digit)
4. Group leaves under each stem
5. Read distribution left-to-right
6. Find min, max, range, median

Formulas:

Stem = floor(x / 10)
Leaf = x mod 10
Range = Max – Min
Median = middle value (sorted)
Excel: MAX(), MIN(), MEDIAN()

No chart — display is the visual itself

3. Box Plot & Dot Plot

Steps:

1. Sort each brand's data
2. Find 5-Number Summary:
   Min, Q1, Median, Q3, Max
3. IQR = Q3 – Q1
4. Outlier fence: < Q1–1.5*IQR
   or > Q3+1.5*IQR
5. Dot plot: mark each value on axis

Formulas:

Q1 = QUARTILE(data, 1)
Q2 = MEDIAN(data)
Q3 = QUARTILE(data, 3)
IQR = Q3 – Q1
Fence = Q1–1.5*IQR to Q3+1.5*IQR

Chart: Grouped bar (5-num) + Scatter Dot

4. Binomial Distribution Fitting

Steps:

1. Find mean: x■ = Σ(x·f) / N
2. Set n = no. of trials (given)
3. Estimate p = x■ / n, q = 1–p
4. P(X=x) = C(n,x)·p■·q^(n–x)
5. Expected = N · P(X=x)
6. Round expected to integers
7. Chi-sq test: compare obs vs exp

Formulas:

P(X=x) = C(n,x) · p^x · (1–p)^(n–x)
Mean = np, Var = npq
C(n,x) = n! / [x!(n–x)!]
χ² = Σ (O–E)² / E
Excel: BINOMDIST(x,n,p,FALSE)

Chart: Clustered bar — Observed vs Expected

5. Poisson Distribution Fitting

Steps:

1. Find mean: λ = Σ(x·f) / N
2. Use Poisson PMF for each x
3. Expected = N · P(X=x)
4. Round to nearest integer
5. Verify ΣExpected = N
6. Chi-sq test goodness of fit

Formulas:

P(X=x) = (e^–λ · λ^x) / x!
λ = Mean = Variance (for Poisson)
e ≈ 2.71828
χ² = Σ (O–E)² / E
Excel: POISSON(x, λ, FALSE)

Chart: Clustered bar — Observed vs Expected

6. Regression Analysis (Y on X)

Steps:

1. Compute ΣX, ΣY, ΣXY, ΣX², n
2. Find b (slope) using formula
3. Find a (intercept) using formula
4. Write line: Yc = a + bX
5. Calculate fitted Yc for each X
6. Compute SSE = Σ(Y–Yc)²
7. Find R and R² (goodness of fit)

Formulas:

b = (nΣXY – ΣX·ΣY) / (nΣX² – (ΣX)²)
a = (Y■ – b·X■) or a = (ΣY – bΣX)/n
Yc = a + bX
R² = 1 – SSE/SST
Excel: SLOPE(), INTERCEPT(), CORREL()

Chart: Scatter plot + red regression line

7. Sample Correlation Coefficient (Pearson's r)

Steps:

1. List Xi and Yi pairs
2. Compute X■ = ΣXi/n, Y■ = ΣYi/n
3. Find σx = STDEV(X), σy = STDEV(Y)
4. Standardise: Zi(X)=(Xi–X■)/σx
5. Standardise: Zi(Y)=(Yi–Y■)/σy
6. Compute Zi(X)·Zi(Y) for each pair
7. r = ΣZi(X)·Zi(Y) / (n–1)

Mathematical Formulas:

r = Σ(Xi–X■)(Yi–Y■) / (n–1)σxσy
r = ΣZi(X)·Zi(Y) / (n–1)
r = [nΣXY–ΣXΣY] /
√[(nΣX²–(ΣX)²)(nΣY²–(ΣY)²)]
–1 ≤ r ≤ +1
Excel: =CORREL(X_range, Y_range)

Interpretation of r:

r = +1 Perfect positive correlation
r = –1 Perfect negative correlation
r = 0 No linear correlation
0.7≤|r|<1 Strong correlation
0.4≤|r|<0.7 Moderate correlation
|r| < 0.4 Weak correlation
Chart: Scatter plot (Xi vs Yi)